Student Full Name: Malibekov Iliyas
Student ID Number: 230103344

Institutional Email / Git URL: 230103344@sdu.edu.kz

Practicum Date & Time (Slot): 25.09.2026

Laptop Model & OS (Linux/macOS/Win): mac

CPU Model & Generation: Apple M3

Physical Cores (P/E breakdown if hybrid): 8

Logical SMT Threads (omp_get_max_threads): 8

Target Workload Size N (Derived via Formula): 4

L1 / L2 Cache Line Size (Typically 64 bytes): 128

Phase 1:

Parameter	Value
Laptop	MacBook Air, Mac15,13
Operating system	macOS 15.2
CPU	Apple M3 (ARM64, 3 nm process)
Physical cores	8 = 4 P-cores + 4 E-cores (hybrid)
Logical CPUs / omp_get_max_threads()	8 / 8 (no SMT)
Cache line size	128 bytes
L1d cache (P-core)	128 KB (131,072 B)
L2 cache (P-cluster, shared)	16 MB (16,777,216 B)

Phase 2:

Measurement	Value
Run 1 (cold, discarded)	1.277623 s
Run 2	1.249203 s

Run 3	1.269286 s
T_seq = (Run 2 + Run 3) / 2	1.259245 s
Run 2 vs Run 3 spread	1.59 %
Maximum stopping time	685
Checksum (Σ steps mod 1,000,000,007)	628244555

Phase 3:

Threads (k)	Run 1 (Cold)	Run 2 (s)	Run 3 (s)	Avg T_k (s)	S_emp(k)	S_theo(k)	Delta (k)
k = 1 (T_1)	1.2765	1.2474	1.2493	1.2484	1.009	1.000	−0.009
k = 2	0.6683	0.6668	0.6660	0.6664	1.890	1.890	0.000
k = 4	0.3652	0.3659	0.3647	0.3653	3.447	3.403	−0.044
k = 8 (max)	0.2206	0.2256	0.2318	0.2287	5.506	5.677	+0.171
k = 16	—	—	—	—	—	—	—

Table 1 — Empirical measurement log (S_emp(k) = T_seq / T_k, T_seq = 1.259245 s)

	k = 1	k = 2	k = 4	k = 8
Parallel efficiency S_emp(k) / k	100.9 %	94.5 %	86.2 %	68.8 %